

Old UNM Algebra Qualifier Problems

Based on Chapters 1 and 2 of Dummit–Foote

Fifteen-year collection: January 2012–August 2023

Scope. This packet uses the third edition of Dummit and Foote, *Abstract Algebra*. Chapter 1 is “Introduction to Groups” and Chapter 2 is “Subgroups.” The first section contains problems that can reasonably be attempted after those two chapters. The second section records closely related problems whose standard solutions use some later material, such as cosets and Lagrange’s theorem, quotient groups, conjugacy classes, or more developed group actions.

I. Core problems from Chapters 1–2

1. January 2012 Qualifier, Problem 3

Prove that the group with generators a, b and relations

$$a^3 = b^2 = e, \quad ab = ba^2,$$

is isomorphic to the symmetric group S_3 .

Solution. Let G denote the presented group. From $ab = ba^2$ we obtain

$$aba = b \implies ba = a^2b.$$

Consequently every word in a and b can be rewritten with all powers of a on the left and with at most one occurrence of b . Using $a^3 = e$ and $b^2 = e$, every element of G is therefore represented by one of

$$e, a, a^2, b, ab, a^2b.$$

Thus $|G| \leq 6$. Define a map $\varphi : G \rightarrow S_3$ given by sending

$$a \mapsto (123), \quad b \mapsto (12).$$

Note that $(\varphi(a))^3 = (\varphi(b))^2 = e$. Moreover,

$$\varphi(a) \cdot \varphi(b) = (123)(12) = (13) = (12)(132) = \varphi(b) \cdot (\varphi(a))^2,$$

We thus have a homomorphism $G \rightarrow S_3$. Its image contains a 3-cycle and a transposition, which generate S_3 ; hence it is surjective. Since $|G| \leq 6 = |S_3|$, this surjection is an isomorphism.

2. August 2012 Qualifier, Problem 2

Prove that the group defined by generators a, b, c, d and relations

$$adb = b^2a, \quad a^3 = c, \quad b^2 = c, \quad d^2 = c$$

is infinite and noncommutative.

Solution. The last three relations give $a^3 = b^2 = d^2 = c$. In the first relation we may replace b^2 by a^3 and obtain

$$adb = a^4.$$

Multiplying on the left by a^{-1} gives $db = a^3 = b^2$, and then multiplying on the right by b^{-1} gives $d = b$. The generators c, d can therefore be eliminated, and the group has the equivalent presentation

$$G \cong \langle a, b \mid a^3 = b^2 \rangle.$$

There is a surjective homomorphism $G \rightarrow \mathbb{Z}$ defined additively by

$$a \mapsto 2, \quad b \mapsto 3,$$

because $3 \cdot 2 = 2 \cdot 3$, and because 2 and 3 generate $\mathbb{Z}$. Thus G is infinite. There is also a surjection $G \rightarrow S_3$ given by

$$a \mapsto (123), \quad b \mapsto (12),$$

since both a^3 and b^2 map to the identity. Because S_3 is noncommutative, G cannot be commutative.

3. January 2014 Qualifier, Problem 1

Let G be a group. Prove that G is abelian if and only if the map

$$f : G \longrightarrow G, \quad f(g) = g^2,$$

is a homomorphism.

Note: Solving this also solves **January 2019 Qualifier, Problem 1**.

Solution. If G is abelian, then for all $x, y \in G$,

$$f(xy) = (xy)^2 = x^2y^2 = f(x)f(y),$$

so f is a homomorphism. Conversely, if f is a homomorphism, then

$$xyxy = (xy)^2 = f(xy) = f(x)f(y) = x^2y^2.$$

Cancel x on the left and y on the right to get $yx = xy$. Since this holds for arbitrary x, y , the group is abelian.

4. January 2014 Qualifier, Problem 2

Show that the diagonal subgroup

$$D = \{(a, a) : a \in S_3\} \subseteq S_3 \times S_3$$

is not normal.

Solution. Take $g = (12)$ and $a = (123)$. Conjugating $(a, a) \in D$ by (g, e) gives

$$(g, e)(a, a)(g, e)^{-1} = (gag^{-1}, a) = ((132), (123)).$$

The two coordinates are different, so this element is not in D .

5. August 2016 Qualifier, Problem 2

Let G be a group. Show that the set of inner automorphisms of G is a normal subgroup of the group of automorphisms of G .

Note: For $g \in G$, define $c_g : G \rightarrow G$ by $c_g(x) = gxg^{-1}$. The set $\text{Inn}(G) = \{c_g : g \in G\}$ is called the set of **inner automorphisms** of G .

Solution. For $g \in G$, write $c_g(x) = gxg^{-1}$. Since

$$c_g \circ c_h = c_{gh}, \quad c_g^{-1} = c_{g^{-1}},$$

the set $\text{Inn}(G) = \{c_g : g \in G\}$ is a subgroup of $\text{Aut}(G)$. If $\alpha \in \text{Aut}(G)$, then for every $x \in G$,

$$(\alpha c_g \alpha^{-1})(x) = \alpha(g \alpha^{-1}(x) g^{-1}) = \alpha(g) x \alpha(g)^{-1} = c_{\alpha(g)}(x).$$

Thus $\alpha c_g \alpha^{-1} \in \text{Inn}(G)$, proving that $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$.

6. January 2018 Qualifier, Problem 7

Suppose that, for some group G , $|\text{Aut}(G)| = 1$. Show that G is abelian and every element of G has order at most 2.

Note: This problem is repeated in **August 2022 Qualifier, Problem 1** and in **August 2023 Qualifier, Problem 2**.

Solution. Every inner automorphism belongs to $\text{Aut}(G)$, so the hypothesis implies that every inner automorphism is the identity. Hence $gxg^{-1} = x$ for all $g, x \in G$, whence $gx = xg$ for all $g, x \in G$. Consequently G is abelian.

In an abelian group the inversion map $\iota(g) = g^{-1}$ is an automorphism. It must therefore be the identity. Thus $g^{-1} = g$, so $g^2 = e$ for every $g \in G$. Every element consequently has order 1 or 2.

7. August 2020 Qualifier, Problem 10

Prove that the group with generators a, b and relations

$$a^2 = b^2 = e$$

is infinite.

Solution. The element ab has infinite order.

8. January 2021 Qualifier, Problem 3

Prove that for every noncyclic subgroup G of $(\mathbb{R}, +)$ and every pair of real numbers $a < b$, there exists $g \in G$ such that $a < g < b$.

Solution. Let $P = G \cap \mathbb{R}_{>0}$. We first show that $\inf P = 0$. Suppose instead that $\delta = \inf P > 0$. Choose elements of P converging to δ . Two distinct sufficiently late terms would have positive difference smaller than δ that also lies in P , contradicting the definition of the infimum. Hence the sequence is eventually constant and $\delta \in P$. For any $x \in G$, write

$$x = q\delta + r, \quad q \in \mathbb{Z}, \quad 0 \leq r < \delta.$$

Then $r = x - q\delta \in G$. Minimality of δ forces $r = 0$, so $G = \delta\mathbb{Z}$, contrary to the assumption that G is noncyclic. Thus $\inf P = 0$.

Choose $h \in G$ with $0 < h < b - a$. There is an integer k satisfying

$$\frac{a}{h} < k < \frac{b}{h},$$

because the interval has length $(b - a)/h > 1$. Then $g = kh \in G$ and $a < g < b$, as required.

9. January 2023 Qualifier, Problem 2

Show that S_A acts on $\mathcal{P}(A)$, the power set of A , via

$$\sigma \cdot B = \{\sigma(b) : b \in B\}.$$

When $A = \{1, 2, 3, 4\}$, determine the following:

- (a) the orbit of $\{1, 2\}$;
- (b) the stabilizer of $\{1, 3, 4\}$;
- (c) the fixed set of $(12)(34)$.

Solution. The identity permutation fixes every subset, and

$$(\sigma\tau) \cdot B = \{\sigma(\tau(b)) : b \in B\} = \sigma \cdot (\tau \cdot B),$$

so this is a group action.

- (a) Permutations preserve cardinality, and S_4 acts transitively on the 2-element subsets. Hence

$$\text{Orb}(\{1, 2\}) = \{\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}\}.$$

- (b) A permutation stabilizes $\{1, 3, 4\}$ precisely when it fixes its complement $\{2\}$. Therefore

$$\text{Stab}(\{1, 3, 4\}) = \{\sigma \in S_4 : \sigma(2) = 2\} \cong S_3.$$

Explicitly, it consists of the six permutations of $\{1, 3, 4\}$, extended by fixing 2.

- (c) A subset is fixed by $(12)(34)$ exactly when it is a union of the cycles $\{1, 2\}$ and $\{3, 4\}$. The fixed subsets are

$$\emptyset, \quad \{1, 2\}, \quad \{3, 4\}, \quad \{1, 2, 3, 4\}.$$

II. Closely related or borderline problems

1. August 2012 Qualifier, Problem 3

Let σ and τ be distinct 3-cycles in S_5 . Show that $\langle \sigma, \tau \rangle$ is either isomorphic to A_4 or is A_5 .

Correction to the printed problem. The assertion is false as printed. If $\sigma = (123)$ and $\tau = (132) = \sigma^{-1}$, then the cycles are distinct but $\langle \sigma, \tau \rangle = \langle \sigma \rangle \cong C_3$. The correct classification is

$$\langle \sigma, \tau \rangle \cong C_3, \quad A_4, \quad \text{or} \quad A_5.$$

Note: Will be solved later.

2. August 2014 Qualifier, Problem 1

A finite group G acts on itself by conjugation. Determine all possible G if this action yields precisely three distinct orbits.

Solution. The orbits are the conjugacy classes. If G is abelian, every class is a singleton, so $|G| = 3$ and $G \cong C_3$.

Assume now that G is nonabelian and set $m = |G|$. We claim $Z(G) = \{e\}$. Every central element is a singleton conjugacy class. Since there are only three classes, a nontrivial center would have order 2; all $m - 2$ noncentral elements would then form one conjugacy class. Its size $m - 2$ must divide m , hence must divide 2. This forces $m = 3$ or 4, for which every group is abelian, a contradiction.

Let the two nonidentity conjugacy classes have sizes $a \leq b$. Then

$$a + b = m - 1,$$

and each of a, b is a proper divisor of m , hence at most $m/2$. If both were smaller than $m/2$, their corresponding centralizers would have index at least 3, so $a, b \leq m/3$, contradicting $a + b = m - 1$ for $m > 3$. Therefore $b = m/2$ and $a = m/2 - 1$. Since $a \mid m$, writing $d = m/2 - 1$ gives $d \mid 2$, so $m = 4$ or 6. The nonabelian case forces $m = 6$, and the unique nonabelian group of order 6 is S_3 . Both C_3 and S_3 indeed have three conjugacy classes. Hence

$$G \cong C_3 \quad \text{or} \quad G \cong S_3.$$

3. August 2014 Qualifier, Problem 2

Let G be the group of matrices

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a \in (\mathbb{Z}/p\mathbb{Z})^\times, b \in \mathbb{Z}/p\mathbb{Z} \right\}.$$

Find all normal subgroups of G .

Note: Will be solved later.

4. August 2016 Qualifier, Problem 1

Prove that if G and H are two finite groups with coprime orders, then there is no nontrivial homomorphism between them.

Solution. Let $\varphi : G \rightarrow H$. The image $\varphi(G)$ is a subgroup of H , so its order divides $|H|$. By the first isomorphism theorem, $|\varphi(G)| = [G : \ker \varphi]$, so it also divides $|G|$. Since $\gcd(|G|, |H|) = 1$, we obtain $|\varphi(G)| = 1$. Thus $\varphi(G) = \{e_H\}$ and φ is trivial.

5. January 2017 Qualifier, Problem 3

Let $K \leq H \leq G$ be a chain of subgroups.

- If H is characteristic in G and K is characteristic in H , show that K is characteristic in G .
- Show that if “characteristic” is replaced by “normal” throughout the preceding statement, then the amended statement is false.

Note: A subgroup $H \leq G$ is **characteristic** if $\alpha(H) = H$ for every $\alpha \in \text{Aut}(G)$.

Solution. For (a), let $\alpha \in \text{Aut}(G)$. Since H is characteristic, $\alpha(H) = H$, and the restriction $\alpha|_H$ is an automorphism of H . Since K is characteristic in H , this restriction preserves K . Therefore $\alpha(K) = K$ for every automorphism α of G , proving that K is characteristic in G .

For (b), take $G = D_4$ and $H = \langle r^2, s \rangle$, $K = \langle s \rangle$.

6. January 2019 Qualifier, Problem 3

Suppose G acts on both X and Y . A map $\varphi : X \rightarrow Y$ is called a G -isomorphism if it is a bijection and

$$g\varphi(x) = \varphi(gx)$$

for every $g \in G$ and $x \in X$. Let G act transitively on X , fix $x_0 \in X$, let H be the stabilizer of x_0 , and let L be the set of left cosets of H . Show that X and L are G -isomorphic.

Solution. Define

$$\Phi : G/H \longrightarrow X, \quad \Phi(gH) = g \cdot x_0.$$

If $gH = g'H$, then $g'^{-1}g \in H$, so $g'^{-1}g \cdot x_0 = x_0$ and hence $g \cdot x_0 = g' \cdot x_0$. Thus Φ is well defined. Transitivity makes it surjective. Conversely, if $g \cdot x_0 = g' \cdot x_0$, then $g'^{-1}g \in H$, so $gH = g'H$; hence it is injective. Finally, for $k \in G$,

$$\Phi(k \cdot gH) = \Phi(kgH) = kg \cdot x_0 = k \cdot \Phi(gH).$$

Therefore Φ is a G -equivariant bijection.

7. August 2019 Qualifier, Problem 4

Suppose M is a maximal proper subgroup of a group G . Prove that either $N_G(M) = M$ or $N_G(M) = G$. Deduce that if M is a maximal subgroup of a finite group G which is not normal, then the number of nonidentity elements contained in the conjugates of M is

$$(|M| - 1)(G : M).$$

Correction to the printed problem. The second part should be as follows: Assume M is a maximal subgroup of a finite group G which is not normal, and distinct conjugates of M intersect only in the identity. Then

$$\left| \left(\bigcup_{g \in G} gMg^{-1} \right) \setminus \{1\} \right| = (|M| - 1)(G : M).$$

Solution. See [Exercise 23, Section 4.3. (solved in Homework 6)].

8. August 2023 Qualifier, Problem 1

Suppose H is a subgroup of a finite group G and

$$\gcd(|H|, (G : H)) = 1.$$

Is H necessarily the unique subgroup of G of order $|H|$? If the answer is yes, give a proof; otherwise, give a counterexample.

Note: The answer is “yes” if H is normal, see [Exercise 19, Section 3.2. (solved in [Homework 5](#))].

Solution. No. Let $G = S_3$ and $H = \langle(12)\rangle$. Then $|H| = 2$ and $[G : H] = 3$, so

$$\gcd(|H|, [G : H]) = \gcd(2, 3) = 1.$$

Nevertheless, S_3 has three distinct subgroups of order 2:

$$\langle(12)\rangle, \quad \langle(13)\rangle, \quad \langle(23)\rangle.$$

Thus the coprimality condition does not imply uniqueness.

Source

University of New Mexico, Department of Mathematics and Statistics, [Past Qualifying Exams – Algebra](#). Individual exam dates and problem numbers are recorded above.